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Applied dynamical systems

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What is a dynamical system? (and examples)

“A dynamical system is a set of variables and their evolution in time.”

More formally, let M be a compact phase space. We call $\varphi^t : M \rightarrow M$ the *dynamical system*. We may enforce $t \in \mathbb{Z}$ (wherein this is called a map) or $t \in \mathbb{R}$ (wherein this is called a flow¹). We say a map φ^t is an *autonomous* map if

$$\varphi^t = \underbrace{\varphi \circ \dots \circ \varphi}_{t \text{ times}}$$

We say φ^t is a *non-autonomous map*² if

$$\varphi^t = \varphi_t \circ \dots \circ \varphi_1.$$

The main moral of all of this distinction is make clear that *dynamics are not differential equations and/or group theory* (in the same sense that what is happening at a point in time is not always fixed, as those subjects can be thought of as autonomous dynamical systems). In general, this definition of a dynamical system is quite loose and thus dynamical systems are a very interdisciplinary subject that draws from many areas of mathematics (e.g., perturbation theory is closely tied to differential topology, probability theory when dealing with randomness).

Example 1. Let $d = 1$ and $M = [0, 1]$, and $\varphi(x) = x$. This is kind of uninteresting because nothing happens.

Non-example 2. Let $d = 1$ and $M = [0, 1]$. Let $\varphi(x) = 2x$. This fails to be a dynamical system because φ does not map M to M .

Example 3. Let

$$\frac{dx}{dt} = v(t, x) \quad x \in \mathbb{R}^d.$$

This is implicitly a non-autonomous dynamical system. This can be rewritten:

$$\frac{d\varphi^t(x)}{dt} = v(t, \varphi^t(x)) \quad x \in \mathbb{R}^d.$$

We think of x as an “initial condition” rather than a position (given by $\varphi^t(x)$).

We call $\{\varphi^t(x)\}_t$ the *orbit* (or trajectory) of x ; dynamical systems is the study of all possible orbits. People are also very interested in perturbations of dynamical systems, but there are many ways and perspectives to look at this.

Definition 4. A *linear* dynamical system can be written as

$$\frac{dx}{dt} = Ax \quad x \in \mathbb{R}^d.$$

Example 5. Let $\varphi(x) = 2x \% 1$, i.e.,

$$\varphi(x) = \begin{cases} 2x & x < 1/2 \\ 2x - 1 & x \geq 1/2 \end{cases}.$$

We call this a *linear hyperbolic map*, but this is an easy example of a nonlinear dynamical system.

¹ Note that we also allow invertible flows/maps here, which may not necessarily always be the case.

² with abuse of notation for φ^t

Example 6. We can think of *next-token prediction* as a (random) dynamical system too. Let X_t be the tokens. Then,

$$\varphi^t(x) \sim P(X_t | X_1, \dots, X_{t-1}).$$

In this example, M is the discrete space of strings. This is a “non-autonomous” dynamical system.

Example 7. The Navier-Stokes equations (which have been in the news) are an autonomous dynamical system. But the solution of the PDE lives in a function space, so the phase space is a function space (i.e. L^2). Here, our ‘ x ’ is the initial velocity (which is a function). We will not study this because M is not compact and is also infinite-dimensional.

Linear systems

Definition 8. A *linear system* is a dynamical system given by

$$\frac{\partial \varphi^t}{\partial t}(x) = A\varphi^t(x)$$

where A is a matrix.

For now, let us assume the solutions exist and are unique. Linear systems are important because they are just about the only systems that have closed-form solutions. In practice, though, nearly everything we write down does not have a closed-form solution and needs to be approximated. Recall that the solution is given by

$$\varphi^t(x) = e^{tA}x,$$

which is further discussed in the book.

We have two motivations for studying linear systems:

- They arise in linear stability analysis / perturbation theory.
- They are kind of basic for analysis.

In contemporary research, we don’t really think about continuous-time dynamics anymore – we just take the discretization as the “ground-truth”. This is just because the systems we look at now are so complicated we can’t hope to garner much about an analytical solution.

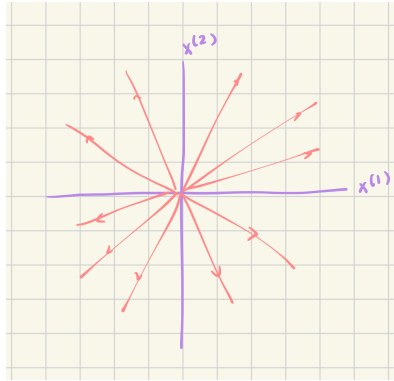
Let us just look at maps instead of ODEs. This case is very easy: the solution to

$$x_{t+1} = Bx_t$$

is given by

$$\varphi^t(x) = B^t x.$$

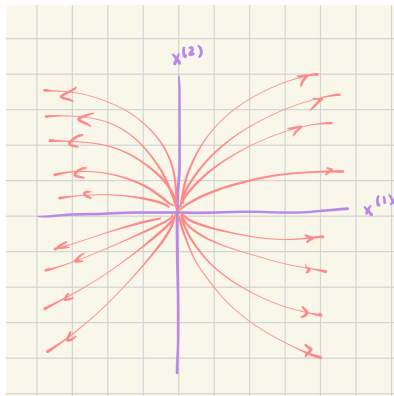
Example 9. Let $B = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$. What is $\varphi^t(x)$? In the case of $\lambda \neq 1$,



. We call the origin a *node* in this case. It is *stable* if $0 < \lambda < 1$ and *unstable* if $\lambda > 1$.

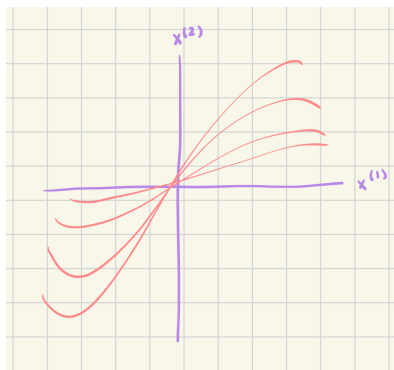
We should think of coordinates $x^{(1)}$ as a “projection” of x onto the standard basis.

Example 10. Let $B = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$. What if $\lambda > \mu > 0$?



Example 11. Let $B = \lambda \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$. We call the origin a *focus*.

Example 12 (Hyperbolic system). Let $B = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$.



We call the origin a *saddle point*. THIS IS WRONG!

These are called phase portraits, but these only make sense for flows (and not maps)! As an exercise, one can prove for each of the exercises that the curve is invariant under the map; i.e., let a curve be written as $x^{(2)} = f(x^{(1)})$. Show that

$$x^{(2)}(\varphi^t(x)) = f(x^{(1)}(\varphi^t(x))).$$

What is the relationship between the map and the flow? Let

$$\begin{aligned}\dot{x} &= Ax \\ x_{t+1} &= Bx_t.\end{aligned}$$

We have discussed the latter map, but what is the continuous time flow when

$$B = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix},$$

then using the solution to the continuous ODE, we can write

$$A = \begin{pmatrix} \log \lambda & 0 \\ 0 & \log \lambda \end{pmatrix}.$$

Thus, we can't study the phase portraits with for *maps* with negative λ (but we certainly can for *flows*!).

Review of linear algebra

In this section, we consider **matrices with entries over the reals**, and briefly review some linear algebra.

When is a (square) matrix A diagonalizable (i.e., $A = P^{-1}\Lambda P$ where Λ is diagonal)? Such a diagonalization exists when the column space spans all of $\mathbb{R}^d$ (or when the eigenvectors span $\mathbb{R}^d$). If A is diagonalizable, it is easy to compute $A^{-1} = P\Lambda^{-1}P^{-1}$. Thus, a matrix and its inverse have the same eigenvectors.

We introduce two notions of multiplicity:

Definition 13. The **algebraic multiplicity** of an eigenvalue λ is the multiplicity of λ with respect to the characteristic polynomial $\det(A - \lambda I) = 0$. An eigenvalue is *simple* if its (algebraic) multiplicity is one.

Definition 14. The **geometric multiplicity** of λ is equal to $\dim(\text{null}(A - \lambda I))$.

If the algebraic multiplicity is equal to the geometric multiplicity for all eigenvalues, then the matrix is diagonalizable. However, in general, the following fact is true:

Fact 15. *The algebraic multiplicity is always greater than or equal to the geometric multiplicity.*

In some ways, the algebraic multiplicity is also geometric (here, we diverge from the book). Let us introduce the **Jordan decomposition**, which is the most general decomposition of a matrix that always exists. It is of the form

$$P^{-1}AP = J,$$

where J has the Jordan block structure

$$J_\lambda = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & \ddots \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

and

$$J = \begin{pmatrix} J_{\lambda_1} & & \\ & J_{\lambda_2} & \\ & & \ddots \\ & & & J_{\lambda_k} \end{pmatrix},$$

where J_λ is a Jordan block.

To interpret the Jordan decomposition, we introduce generalized eigenvectors of A .

Definition 16. We say v is a **generalized eigenvectors** of A if, for some $k > 0$, $(A - \lambda I)^k v = 0$. Thus, the generalized eigenvalue satisfies $\det(A - \lambda I)^k = 0$.

As an example, let us consider $k = 2$: If

$$(A - \lambda I)(A - \lambda I)v_2 = 0$$

and

$$(A - \lambda I)v_1 = 0,$$

then (check this)

$$(A - \lambda I)v_2 = v_1,$$

so we can relate the $k = 2$ generalized eigenvectors to the eigenvectors of A . We call v_1, v_2 a *chain*. It is a deep theorem that the span of the generalized eigenvalues $\{v_1, \dots, v_d\}$ span $\mathbb{R}^d$, which is equivalent to the fact that a Jordan decomposition exists. Call E^λ the *generalized eigenspace* associated with λ .

Fact 17. E^λ is invariant under the map and the flow (i.e., for flows, if $x \in E^\lambda$, then $e^{tA}x \in E^\lambda$).

This is the reason we discuss the Jordan form. Some fast facts:

Fact 18. The number of Jordan blocks J_λ for that eigenvalue is equal to the geometric multiplicity of that eigenvalue.

Fact 19. The size of the Jordan block is how many eigenvectors we can chain. For given λ , the sum of the sizes of the Jordan blocks J_λ is equal to the algebraic multiplicity.³

Now, when A is diagonalizable, let us reconsider the general solution to

$$\frac{\partial \varphi^t(x)}{\partial t} = A\varphi^t(x)$$

³ There is some discrepancy in that the book introduces algebraic multiplicity as something like the size of the largest Jordan block

is

$$\varphi^t(x) = \sum_{i=1}^d c_i v_i e^{\lambda_i t}.$$

It is simple to verify that this is a solution just by taking the derivative.

More on linear systems

Let us rediscuss the issue of invertibility. It is easy to tell when a map $x_{t+1} = Ax_t$ is invertible just by looking at the matrix A . It is more complicated for a flow. Recall again the general solution of a linear flow $\varphi^t(x) = e^{tA}x$, and the following facts:

Definition 20.

$$e^{tA} = I + tA + \frac{t^2 A^2}{2!} + \dots$$

Fact 21. If $\|A\| < \infty$, then $\|e^{tA}\| < \infty$.

Proof. In the book. □

Fact 22.

$$e^{tA}A = Ae^{tA}.$$

Proof. This is easy to see because the powers of a matrix commute (i.e., $AA^n = A^nA$). □

One may now verify

$$\frac{\partial \varphi^t(x)}{\partial t} = e^{tA}Ax = Ae^{tA}x = A\varphi^t(x).$$

We have not assumed anything about A (other than it is bounded). This solution is unique via the following short proof. If $y(t)$ is a solution, then one can compute

$$\frac{\partial}{\partial t} (e^{-tA}y(t)) = 0,$$

so $y(t) = e^{tA}c$, and $y(0) = c$ implies $x = c$. Thus, $y(t) = e^{tA}x$.

How do we compute e^{tA} ? This is discussed extensively in the book.

Suppose A is diagonalizable, so $A = P^{-1}\Lambda P$, and $e^{tA} = Pe^{t\Lambda}P^{-1}$. Note now that

$$e^{t\Lambda} = (e^{\lambda_1 t} \dots).$$

An equivalent way of expressing this is $\varphi^t(x) = \sum_{i=1}^d c_i v_i e^{\lambda_i t}$.

Let us suppose A is not diagonalizable. Then, we can write $e^{tA} = Pe^{Jt}P^{-1}$, where the columns of P are generalized eigenvectors. What is e^{Jt} ? Suppose we have a Jordan block of size 2 with generalized eigenvectors v_1, v_2 , and corresponding

$$J_\lambda = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

The action of A on some v_1 is the action of J_λ on $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, which evaluates to $\lambda \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. This does not really say anything interesting because the first generalized eigenvector v_1 of the chain is just the regular eigenvector. So now, we consider

$$J_\lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ \lambda \end{pmatrix}.$$

In Euclidean coordinates, this is

$$J_\lambda v_2 = v_1 + \lambda v_2,$$

which is exactly what we did earlier.

As an exercise, compute:

Fact 23.

$$e^{tJ_\lambda} = \begin{pmatrix} e^{t\lambda} & t \\ 0 & e^{t\lambda} \end{pmatrix},$$

so we get $e^{t\lambda}v_1 + te^{t\lambda}v_2$. Can we extend this to $d \times d$ matrices?

References